

CONTACT

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GITHUB hsandmann
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ORCID 0000-0001-7758-695X
PROJECT graphole.com
LOCATION São Paulo, Brazil
RELOCATION Open to relocation (on-site)

SKILLS

Languages

Java · Python · Go · Kotlin · C · C++ · JavaScript · SQL

Machine Learning & AI

Deep Learning · Neural Networks · Computer Vision · NLP · PCA · SVM · Clustering

Backend & Architecture

Spring Boot · Spring Cloud · Microservices · Design Patterns · Software Architecture

Data & Infrastructure

PostgreSQL · Redis · Kafka · RabbitMQ · Docker · Kubernetes · Spark · MongoDB · Neo4j

Domains

Banking & Fintech · Cryptography (PKI/HSM) · High-Performance Computing · POS/ERP

EDUCATION

D.Sc., Electrical Engineering
Polytechnic School, USP 2007–2012

M.Sc., Electrical Engineering
Polytechnic School, USP 2003–2006

B.Sc., Computer Science
FEI 1999–2002

LANGUAGES

Portuguese	Native
English	Advanced
German	B1

Humberto Sandmann

Senior Software Architect & AI Engineer

PROFILE

Senior software architect and artificial intelligence engineer with 25+ years building high-stakes systems across banking, fintech, and retail — core banking, cryptography (PKI/HSM), high-performance computing, computer vision, and cloud-native microservices. D.Sc. in neural computing; also a university professor teaching software design, AI, and distributed systems.

EXPERIENCE

Founder & Consulting IT Specialist · H-Sandmann — 2006–2025

Independent consultancy delivering short-term, specialized engagements in machine learning, optimization and security.

Machine Learning · Software Architecture

Co-Founder & Research Scientist · SelsanTech — 2012–2023

Co-founded a venture bringing computational-intelligence research to market; built ZigTrack, real-time indoor tracking of equipment with low-cost fixed sensors.

Machine Learning · Software Architecture

Senior Software Architect & Machine Learning Engineer · RockSpoon — 2018–2019

Team leader and product owner; architected POS/ERP/payroll platforms and computer-vision/ML features for a restaurant-tech company.

POS · ERP · JavaEE · PostgreSQL · OpenCV · Computer Vision · Go · Android · Kotlin

Senior Consultant, Computational Optimization · Banco Votorantim — 2015–2016

Return-on-equity computation for 2M+ records within a 12-hour window on high-performance computing infrastructure for a major bank.

RoE · High-Performance Computing · J2EE · Spring Boot · Jenkins · Test Automation

Senior Consultant, Computational Optimization · TecBan — 2011

Senior Software Engineer / Architect · ABN Amro Bank — 2006–2007

Senior System Analyst · Politec — 2005–2006

Senior Software Engineer / Architect · EverSystems — 2004–2005

IT Project Manager · Central Online — 2002–2003

Senior System Analyst · Datamace — 1999–2002

Computer Programmer · Heraeus Electro-Nite — 1998–1999

ACADEMIC POSITIONS

Associate Professor · Insper — 2018–present

Associate Professor II · ESPM — 2014–2026

Associate Professor (freelance) · Link School of Business — 2024

Professor · IBTA — Instituto Brasileiro de Tecnologia Avançada — 2016–2017

Associate Professor · FIAP — 2012–2015

Visiting Professor · Toledo Prudente Centro Universitário — 2015

Associate Professor · Universidade Cruzeiro do Sul — 2012–2013

Temporary Professor · Universidade Nove de Julho — 2011

Teaching Assistant · Polytechnic School, University of São Paulo — 2008–2011

Full academic CV, publications and teaching record at hsandmann.github.io Updated October 2026